

MEMORY LAB — Understanding Linear Attention

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Central claim. Linear attention reorganizes contextual computation so that history can be accumulated into a fixed-shape recurrent state rather than materializing the full token-to-token attention score structure. This can reduce sequence-length-dependent memory pressure during inference, but the compact state introduces retrieval sharpness, capacity and interference trade-offs.

1. Problem and design pressure

In standard self-attention the relationships among tokens are represented by an $N \times N$ score structure. During autoregressive inference prior keys and values are retained in a growing KV cache, so memory usage and data movement increase with context length. The practical bottleneck is therefore not only arithmetic complexity: long-context inference also stresses GPU high-bandwidth memory. Memory Lab deliberately separates two ideas that are often conflated—the quadratic structure of the attention-score matrix and the linear growth of the autoregressive KV cache.

2. What changes technically

Linear-attention methods replace the non-linear softmax with a feature-map factorization $\text{sim}(q,k) = \phi(q) \cdot \phi(k)$. Because the resulting products are associative, the computation can be reordered: key/value contributions are accumulated first into a recurrent state $S_t = S_{t-1} + \phi(k_t) v_t$ (with a corresponding normalizer accumulator). A query then reads the current state. The exact feature map, gating and normalization vary across published variants; the educational substrate visualizes one transparent low-dimensional formulation so that every update remains visible.

3. What the artifact makes observable

Memory Lab is a live educational substrate. A learner can change sequence length N and watch a simplified pairwise footprint grow, run the 100-token race, step through token $\rightarrow$ feature transform $\rightarrow$ outer-product write $\rightarrow$ accumulated state $\rightarrow$ query readout, inspect the state matrix, and test a prediction before the computed result is revealed. All numerical updates are deterministic JavaScript linear algebra; animations are either driven by those computations or explicitly labelled as pedagogical visualization. The toy state is not a production benchmark or an official BDH run.

4. Landscape and comparison

Dimension	Softmax Attention	Linear / Kernel Attention	BDH / BDH-CQ (Pathway)
Decoding memory	$O(N)$ growing KV cache	$O(1)$ fixed $d \times d$ state	$O(1)$ synaptic recurrent state
Core update	Full QK^T then softmax	Rank-1 outer-product write into S	Hebbian / synaptic write into associative memory
Test-time adaptation	Context / fine-tuning	In-state accumulation	Additive demonstration memory (BDH-CQ)
Key trade-off	Sharp retrieval, growing memory	Constant memory, flatter retrieval	Sparse non-negative activity + finite capacity

Related lines of work include CosFormer (2022), RetNet (2023) and Gated Linear Attention (2024). These share the goal of sub-quadratic or constant-memory sequence modelling but differ in feature maps, gating and hardware mapping; they are not interchangeable implementations.

5. BDH / BDH-CQ connection

Pathway’s Dragon Hatchling (BDH) treats memory and reasoning as one computational fabric. In its conceptual model, attention becomes synaptic memory updated by local Hebbian-style writes—the same algebraic outer-product form used by linear attention. BDH-GPU is a GPU-oriented ReLU-low-rank formulation that uses linear attention; the brief explicitly warns against classifying BDH as a Mamba-style SSM. BDH-CQ further relates contextual memory to attention, fast-weight and linear-attention views, with a special case of additive accumulation of demonstration state and no parameter updates at inference. Memory Lab therefore presents the chain: linear attention $\rightarrow$ incremental associative state $\rightarrow$ synaptic / fast-weight perspective $\rightarrow$ BDH & BDH-CQ. It does **not** claim that the toy matrix is an official BDH state or that the browser demo reproduces BDH training or inference.

6. Evidence, maturity and limitations

Recent primary papers establish that linear and recurrent attention remain active research directions. Asymptotic scaling alone does not guarantee lower wall-clock latency; kernels, bandwidth, dimensions and implementation quality matter. Pathway materials report early pretraining-scale experiments (including large parameter counts), but these are developer-reported results rather than independent reproductions. The central educational trade-off remains information capacity: a fixed-shape state can process long sequences without allocating a new pairwise structure, yet finite state produces interference, retrieval degradation and forgetting. Memory Lab exposes this trade-off rather than presenting constant-size memory as lossless.

Primary sources: Qin et al., CosFormer (ICLR 2022); Sun et al., Retentive Network (2023); Yang et al., Gated Linear Attention (ICML 2024); Katharopoulos et al., Transformers are RNNs (2020, foundational); Pathway Dragon Hatchling / BDH-CQ technical materials (2026).

Submission artifact: Public, sign-in-free interactive web explainer (sequence scaling, live state updates, prediction challenge, BDH context). Full source, reproducibility notes, AI disclosure and asset/license record are included in the package.